Problem Set 6 — GAMs and GAMMs

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Sessions 23–27 and Chapters 16–19 of the class notes (GAMs, tensor products, GAMMs). Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Show derivations, give numbers to 3–4 significant digits, and interpret every number you report in context. Use `set.seed(2027)` whenever you simulate. Total: 100 points (estimated effort 6–8 hours).

Notation as in the course: additive predictor $\eta = \beta_0 + \sum_j f_j(x_j)$, penalised criterion $\|\boldsymbol{y} - B\boldsymbol{\beta}\|^2 + \lambda \boldsymbol{\beta}^\top S \boldsymbol{\beta}$, effective degrees of freedom $\mathrm{edf} = \mathrm{tr}(F)$, mixed model $\boldsymbol{y} = X\boldsymbol{\alpha} + Z\boldsymbol{b} + \boldsymbol{\varepsilon}$ with $\boldsymbol{b} \sim N(\boldsymbol{0}, \sigma_b^2 I)$ and $\lambda = \sigma^2/\sigma_b^2$.

```
library(mgcv); library(nlme); library(splines)
```

1 Backfitting, identifiability and dependence between covariates (8 points)

Eight observations are classified by two binary covariates $x_1 \in \{a, b\}$ and $x_2 \in \{u, v\}$:

cell (x_1, x_2)	(a, u)	(a, v)	(b, u)	(b, v)
responses y	10, 12, 14	20	6	14, 16, 18

Fit the additive model $y = \beta_0 + f_1(x_1) + f_2(x_2) + \varepsilon$ with the smoothers S_1 and S_2 equal to *averages within the levels* of x_1 and x_2 respectively (the simplest linear smoothers), and with the constraints $\sum_i f_1(x_{1i}) = \sum_i f_2(x_{2i}) = 0$.

(a) Show that without the constraints the model is not identifiable, and that with them $\hat\beta_0 = \bar y$. Give $\bar y$.

(b) Start the backfitting algorithm at $f_2^{(0)} \equiv 0$ and perform one full sweep ($f_1 \leftarrow S_1(y - \bar y - f_2)$, centre, then $f_2 \leftarrow S_2(y - \bar y - f_1)$, centre). Report $f_1(a), f_1(b), f_2(u), f_2(v)$.

(c) Code the covariates as $z_1, z_2 \in \{-1, +1\}$. Compute $\mathrm{Corr}(z_1, z_2)$ over the eight observations. The error of f_1 is multiplied by $\mathrm{Corr}(z_1, z_2)^2$ in every sweep. Use this to give the limits of $f_1(a)$ and $f_2(u)$ (the iterates after sweeps 1–3 are $f_1(a) = 0.25,\ 1.9375,\ 2.3594$), and check against the least-squares solution.

(d) What would happen to the algorithm and to the model if $\mathrm{Corr}(z_1, z_2)$ were ± 1? Relate this to concurvity.

2 A penalised fit by hand and its mixed-model form (12 points)

Let $n = K = 4$, $B = I_4$, $t = (1, 2, 3, 4)^\top$, $y = (1, 4, 2, 3)^\top$, and the second-order difference penalty $S = D^\top D$ with $D = \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{pmatrix}$. The penalised fit is $\hat{\boldsymbol{\beta}} = (I + \lambda S)^{-1} y$.

(a) Show that S has eigenvalues 10, 2, 0, 0, with orthonormal eigenvectors $u_4 \propto (1, -3, 3, -1)^\top$, $u_3 = \tfrac{1}{2}(1, -1, -1, 1)^\top$ for the two positive eigenvalues, and give a basis of the null space. (Hint: $DD^\top = \begin{pmatrix} 6 & -4 \\ -4 & 6 \end{pmatrix}$.)

(b) Write $\hat{\boldsymbol{\beta}}$ as the sum of the projection of y on the null space and shrunken components along u_3, u_4. Compute $\hat{\boldsymbol{\beta}}$, the residual sum of squares and $\mathrm{edf}(\lambda) = \mathrm{tr}\{(I + \lambda S)^{-1}\}$ for $\lambda = 1$.

(c) Give $\mathrm{edf}(\lambda)$ as a function of λ and its limits at $\lambda \to 0$ and $\lambda \to \infty$. Show that edf $= 3$ exactly at $\lambda = 1/\sqrt{20}$.

(d) Write the fit as a mixed model $y = X\boldsymbol{\alpha} + Z\boldsymbol{b} + \boldsymbol{\varepsilon}$ with $X = [\mathbf{1}, t]$ and $Z = U_+\Sigma_+^{-1/2}$ ($U_+ = [u_3, u_4]$, $\Sigma_+ = \mathrm{diag}(2, 10)$). Show that the BLUP satisfies $\tilde{b}_k = \sqrt{\sigma_k}\, c_k/(1 + \lambda\sigma_k)$ with $c_k = u_k^\top y$, and that $Z\tilde{\boldsymbol{b}}$ equals the penalised component of $\hat{\boldsymbol{\beta}}$. If $\hat{\sigma}^2 = 0.25$ and $\hat{\sigma}_b^2 = 0.25$, what is λ and which component is shrunk more?

3 Tensor-product smooths: construction and units (12 points)

(a) Take the margins $a(x) = (1 - x,\ x)$ and $c(z) = (1 - z,\ z)$ on $[0, 1]$ (K = L = 2) and $f(x, z) = \sum_{k,l} \beta_{kl} a_k(x) c_l(z)$ with $\boldsymbol{\beta} = (\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22}) = (1, 3, 5, 2)$. Write the model-matrix row at $(x, z) = (0.25, 0.8)$ as a Kronecker product and compute $f(0.25, 0.8)$. What kind of surface is this?

(b) With $S_x = S_z = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ write the tensor penalty $\lambda_x \boldsymbol{\beta}^\top (S_x \otimes I_2)\boldsymbol{\beta} + \lambda_z \boldsymbol{\beta}^\top (I_2 \otimes S_z)\boldsymbol{\beta}$ in terms of the β_{kl}, and find the dimension of its null space and a basis for it.

(c) Let $x' = cx$ be a change of units. Show (i) that the marginal penalty $\int g''(x)^2 dx$ of a one-dimensional smooth is multiplied by c^{-3}, so that a tensor-product fit is unchanged if λ_x is replaced by $c^3\lambda_x$; and (ii) that for the isotropic thin-plate penalty $\iint (f_{xx}^2 + 2f_{xz}^2 + f_{zz}^2)\, dx\, dz$ the three terms scale with different powers of c.

(d) *(R)* For the data set `trees` (31 black cherry trees) fit `Volume` (cubic feet) with `family = Gamma(link = "log")` and `method = "REML"` using `s(Girth, Height, k = 10)` and `te(Girth, Height)`, first with `Girth` in inches and `Height` in feet, and again after converting to cm and metres. Report the four AICs and edf, and explain the result in terms of (c).

4 Interaction or not? Fuel economy of 32 cars (16 points)

The data set `mtcars` gives fuel economy `mpg` and, among other variables, weight `wt` (1000 lb) and gross horsepower `hp` of $n = 32$ cars. The two covariates are positively correlated.

(a) Fit by REML: the linear model `lm(mpg ~ wt + hp)`, the additive GAM `s(wt) + s(hp)`, the decomposition `s(wt) + s(hp) + ti(wt, hp)`, and the tensor `te(wt, hp)`. Tabulate AIC, the df used by `AIC()` (from `logLik()`) and the sum of the edf. Which model do you prefer, and by how much? What do the edf of the three smooth terms in the `ti` model tell you about the shape of the effects?

(b) `summary()` of the `ti` model gives $p = 0.001$ for `ti(wt, hp)`, but `anova(am, bm, test = "F")`, with `am`, `bm` the additive and `ti` models fitted with `method = "ML"`, gives $p \approx 0.6$ and a *negative* `Df`. Reproduce both and explain, in a few sentences, why the `anova` result is not a valid test here. Which evidence would you use?

(c) Compute `concurvity()` for the additive model. Interpret the three rows for `s(wt)` and relate them to the correlation of `wt` and `hp`. What can be said safely about the *separate* shapes of f_{wt} and f_{hp}?

(d) Using `predict(..., type = "lpmatrix")` on the `ti` model, estimate the change in mean `mpg` when `hp` rises from 100 to 200 at `wt` $= 2$, 3 and 4, with 95% intervals. Then estimate the *difference* between the changes at `wt` $= 2$ and `wt` $= 4$ with its standard error. Report in words.

(e) Fit `lm(mpg ~ wt * hp)` and compare it with the `ti` model (AIC, coefficients). Write one sentence on what the GAM added or confirmed.

5 Smoothing parameters as variance ratios (12 points)

The data frame `cars` (package `datasets`) has $n = 50$ pairs (`speed` in mph, stopping distance `dist` in ft). We fit `dist` as a smooth function of `speed` with a P-spline: cubic B-splines on 8 equal segments of the range of `speed` (so $K = 11$) and a second-order difference penalty.

(a) Build B (with `splineDesign`), $S = D_2^\top D_2$ and its eigen-decomposition. How many eigenvalues are positive and how many zero? Build $X = [\mathbf{1}, \texttt{speed}]$ and $Z = BU_+\Sigma_+^{-1/2}$ and give their dimensions.

(b) Fit the P-spline by REML with `gam(y ~ B - 1, paraPen = list(B = list(S)), method = "REML")` and the mixed model with `lme(y ~ x, random = list(g = pdIdent(~ Z - 1)), method = "REML")` (one group `g`). Report $\hat{\lambda}$, $\hat{\sigma}^2$, $\hat{\sigma}_b^2$, $\hat{\sigma}^2/\hat{\sigma}_b^2$ and the edf, and the largest difference between the two fitted curves.

(c) Refit the `gam` version with `method = "ML"` and with `method = "GCV.Cp"`. Tabulate $\hat{\lambda}$ and edf for REML, ML and GCV. Explain, using the mixed-model view, why ML might be expected to smooth more than REML, and then run a small simulation (100 data sets from a smooth curve plus noise) to check how large the ML–REML difference in edf typically is.

(d) Compare with the quadratic `lm(dist ~ speed + I(speed^2))`: its fitted values and number of parameters, and a physical reason why stopping distance may be close to quadratic in speed. What does the edf of the P-spline say?

6 Concurvity by hand and by simulation (10 points)

(a) *(by hand)* Let $n = 4$ and let the other terms of a model span $\mathcal{M} = \text{span}\{\mathbf{1}, t\}$ with $t = (1, 2, 3, 4)^\top$. A smooth term has the model-matrix columns $q_1 = (-3, -1, 1, 3)^\top$ and $q_2 = (1, -1, -1, 1)^\top$. Compute (i) $\|P_{\mathcal{M}} q_1\|^2 / \|q_1\|^2$ and the same for q_2; (ii) the *worst-case* concurvity (largest ratio over all coefficient vectors $\boldsymbol{\beta}_j$), (iii) the *observed* concurvity for $\boldsymbol{\beta}_j = (1, 1)^\top$ and (iv) the *estimate* $\|P_{\mathcal{M}} X_j\|_F^2 / \|X_j\|_F^2$, with $X_j = [q_1, q_2]$. Which coefficient vectors make the ratio 1 and 0?

(b) *(R)* Simulate $n = 300$ points: $x_1 \sim U(0, 1)$, $x_2 = \text{clip}(x_1 + N(0, s^2))$ to $[0, 1]$, $y = \sin(2\pi x_1) + 2(x_2 - 0.5)^2 + N(0, 0.3^2)$ with `set.seed(2027)` before each data set. For $s \in \{0.5, 0.1, 0.03\}$ fit `gam(y ~ s(x1) + s(x2), method = "REML")` and report $\text{Corr}(x_1, x_2)$, the `estimate` concurvity of `s(x1)`, the mean standard error of the $s(x_1)$ term and the standard error of the *total fitted value* averaged over observations. Interpret.

7 GAMM for the growth of rats (16 points)

The data frame `nlme::BodyWeight` gives the weight (g) of 16 rats measured on 11 occasions (days 1, 8, 15, …, 64, plus day 44), with three diets (rats 1–8 on diet 1, rats 9–12 on diet 2, rats 13–16 on diet 3). We use `Time` (days), `occ` (the occasion number $1, \ldots, 11$) and `Rat` (as an unordered factor).

```
bw <- as.data.frame(nlme::BodyWeight)
bw$Rat <- factor(as.character(bw$Rat)); bw$Time <- as.numeric(bw$Time)
bw$occ <- as.integer(factor(bw$Time))
c(n = nrow(bw), rats = nlevels(bw$Rat), occasions = max(bw$occ))
```

```
#>         n      rats occasions
#>       176        16        11
```

(a) Draw the 16 growth curves (weight against `Time`, one line per rat, colour by `Diet`). Describe what you see.

(b) Fit with `gam()` and `method = "REML"`: M1 `weight ~ Diet + s(Time, k = 6) + s(Rat, bs = "re")`; M2 = M1 plus `s(Rat, Time, bs = "re")`; M3 = M2 with `s(Time, k = 6)` replaced by `s(Time, by = Diet, k = 6)`. Report AIC, $\hat{\sigma}$ and the within-rat lag-1 correlation of residuals. Which random-effect structure is needed, and why does M3 differ from M2?

(c) Fit with `gamm()` the model `weight ~ Diet + s(Time, by = Diet, k = 6)` with `random = list(Rat = ~ 1 + Time)` (G1), the same with `correlation = corAR1(form = ~ occ | Rat)` (G2), and with random intercept only `random = list(Rat = ~ 1)` and `corAR1` (G3). Compare G1, G2, G3 by AIC (from `$lme`) and report $\hat{\phi}$ and the standard deviation of the random intercept in G3. Explain what happens in G3.

(d) From G1: the standard deviation of the random intercept is 36.96 g, of the random slope 0.2504 g/day, their correlation -0.151 and $\hat{\sigma} = 3.958$ g. Compute the standard deviation of the rat-specific deviation from the population curve at day 1 and day 64, and the ICC of a measurement at day 1. Is the correlation between intercept and slope identifiable from 16 rats? (Give the 95% interval from `intervals(G1$lme)`.)

(e) Summarise in four sentences what the model says about the effect of the diets on growth.

8 Conditional versus population-average; critique and communicate (14 points)

(a) *(derivation)* In a Poisson GAMM, $\log \mu_{ij} = f(x_{ij}) + b_i$ with $b_i \sim N(0, \sigma_b^2)$, $\sigma_b = 0.8$. (i) Derive $\mathrm{E}(y_{ij})$ in terms of $f(x_{ij})$ and σ_b^2. (ii) A typical animal ($b_i = 0$) has conditional mean 5 at $x = 2$: what is the population-average mean? (iii) In a logistic GAMM with $\sigma_b = 2$ and conditional slope 1, approximately what slope should a population-average analysis find?

(b) *(critique)* A student reports on the `mtcars` analysis of Problem 4:

> "We fitted `s(wt) + s(hp) + ti(wt, hp)` by REML. The interaction is significant ($p = 0.001$), so the *effect of weight on fuel economy depends on horsepower*. The additive model has the lower residual deviance, but an `anova` gave $p = 0.59$, so the interaction is not needed. Since `s(hp)` has edf 1.1 it is essentially linear and horsepower is unimportant. The worst-case concurvity of `s(wt)` is 0.96 and so weight is not related to fuel economy."

Identify at least four errors or unsupported statements and write a corrected version.

(c) *(communicate)* In about 150 words, explain to a car buyer (no statistics) what the analysis of Problem 4 says about horsepower, weight and fuel economy, including one sentence on the uncertainty.